1.1.1 This Specification

This booklet contains the specification for MEI Structured Mathematics for teaching from September 2013. It covers Advanced Subsidiary GCE (AS) and Advanced GCE (A Level) qualifications in Mathematics and Further Mathematics and also in Pure Mathematics.

This specification was developed by Mathematics in Education and Industry (MEI) and is assessed by OCR. Support for those delivering the specification comes from both bodies and this is one of its particular strengths.

This specification is designed to help more students to fulfil their potential by taking and enjoying mathematics courses that are relevant to their needs post-16. This involves four key elements: breadth, depth, being up-to-date and providing students with the ability to use their mathematics.

- Most students at this level are taking mathematics as a support subject. Their needs are almost as diverse as their main fields of study, and consequently this specification includes the breadth of several distinct strands of mathematics: Pure Mathematics, Mechanics, Statistics, Decision Mathematics and Numerical Analysis.
- There are, however, those students who will go on to read mathematics at university and perhaps then become professional mathematicians. These students need the challenge of taking the subject to some depth and this is provided by the considerable wealth of Further Mathematics units in this specification, culminating in *FP3*, *M4* and *S4*.
- Mathematics has been transformed at this level by the impact of modern technology: the calculator, the spreadsheet and dedicated software. There are many places where this specification either requires or strongly encourages the use of such technology. The units *DC*, *NC* and *FPT* have computer based examinations; an option in *FP3* is based on graphical calculators, and the coursework in *C3* and *NM* is based on the use of suitable devices.
- Many students complete mathematics courses quite well able to do routine examination questions but unable to relate what they have learnt to the world around them. This specification is designed to provide students with the necessary interpretive and modelling skills to be able to use their mathematics. Modelling and interpretation are stressed in the papers and some of the coursework and there is a comprehension paper as part of the assessment of *C4*.

MEI is a curriculum development body and in devising this specification the long term needs of students have been its paramount concern.

This specification meets the requirements of the Common Criteria (QCA, 1999), the GCE Advanced Subsidiary and Advanced Level Qualification-Specific Criteria (QCA, 1999) and the Subject Criteria for Mathematics (QCA, 2002).